LESSON 13.1
EFFECTS OF OUTLIERS

LESSON INTRODUCTION

MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.

LEARNING TARGETS

- I can determine the implications an outlier has in a given data set.

BENCHMARK COHERENCE

BUILDING ON

MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range.

MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.

WORKING TOWARD

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

ESSENTIAL QUESTIONS

- What is an outlier?
- What measures of center and variation can be used to describe sets of numerical data?
- How does removing an outlier from a data set effect the measures of center and variation?

LESSON NARRATIVE

In this first lesson of Unit 13, students review the concept of outliers and consider their impact on data sets. This lesson begins with a review of the measures of center and variation students have learned in previous grades: mean, median, range, and interquartile range. Students review displays of data and apply what they know to find the measures of center and variation before considering outliers, data points that are far away from the rest in a data set. Students then explore how outliers impact the measures of center and variation.

COMPONENT SUMMARY

13.1.1 WARM-UP *(~5 MIN.)*

Watch a video and reflect on the impact of errors

13.1.2 REFRESHER *(10-15 MIN.)*

Review measures of center, variation, and types of data displays

13.1.3 GUIDED INSTRUCTION *(10 MIN.)*

Intro to stem-and-leaf plots

13.1.4 COLLABORATION *(10-15 MIN.)*

Use a line plot to answer questions about data and measures of center and variation

13.1.5 YOUR TURN! *(5 MIN.)*

Analyze how an outlier impacts a data set

13.1.6 WRAP-UP *(~5 MIN.)*

Reflect on the lesson's learning target

RESOURCES

GLOSSARY

Key Terms:

- Measures of center
- Measure of variation
- Mean
- Range
- Interquartile range
- Median
- Outlier
- Stem-and-leaf plot

Additional Terms: N/A

MATERIALS

- Career Profile video
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- 13.1.1 Warm-Up contains a video to accompany the question prompts.
- (Optional) If pairing students for 13.1.4 Collaboration, consider creating partners ahead of time

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to draw conclusions about the effect of outliers on measures of center and variation.
- Students may think that removing outliers will always have the same impact on the data set, regardless of whether the outlier falls above or below the norm in the set.

THINGS TO CONSIDER

- Determine whether your students need the support of all or part of the refresher.
- If time is an issue, then consider assigning Your Turn! for homework.
- Consider having students refer to the line plot to see the impact of outliers on the shape, center, and spread of the data set by covering the outlier with their hand.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

13.1.2 Refresher - Implement a Notice and Wonder routine to position students to begin to conceptually consider the concept of outliers, which will be reviewed at the end of the component. This routine builds conceptual understanding and encourages synthesis of the information rather than rote memorization of terms.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Throughout this lesson, students will be working to evaluate data that represents real-world contexts, allowing them to connect mathematics to everyday scenarios. This builds investment in the lesson, as well as conceptual understanding of the content.

DIFFERENTIATE INSTRUCTION

13.1.4 Collaboration - Intentionally pair students of varying proficiency levels together, particularly keeping in mind reading ability as this component is reading-comprehension heavy.

13.1.2 Refresher - Consider displaying a graph of data, such as a dot plot, if students are struggling with the idea of center and variation. Visualizing the data may help students understand what center and variation mean.

13.1.5 Your Turn! - Consider providing a visual entry point for this component by allowing students to draw a diagram instead of providing a written explanation for question 1. This will benefit students who may struggle with writing and still allows for an accurate assessment of understanding.

13.1.1 WARM-UP: PRECISION IS KEY

ENRICHMENT

This Warm-Up positions students to consider the importance of precision when working with numbers in mathematics. To connect to the rest of the lesson, ask questions like:

- Would rounding numbers to the wrong decimal place ever impact the correctness of your answer?
- When thinking about answers would you ever say that 141.3 phone calls were made?
- When speaking of money is it accurate to write $5.6?

13.1.1 Warm-Up: Precision Is Key

Becky Miller is a quality control engineer with GE Aviation. Like many other careers, manufacturing engineering has many moving parts that make up an entire process. Becky says that manufacturing is problem-solving.

1. List at least two steps that you take when faced with a difficult task.
   Answers will vary. Understand the problem. Make a plan. Keep trying.

Becky mentions that an error "about the size of a human hair" can make a difference in manufacturing jet engines.

2. What is a time that you have made a small error that caused a big impact in your next steps?
   Answers will vary. I used the wrong inverse operation when solving an equation. I woke up late for school.

13.1.2 Refresher: Measures of Center and Variation

Measures of center, such as **mean** and **median,** are used to describe a typical value in the data set.

Measures of variation, such as **range** and **interquartile range,** are used to describe how "spread out" the data is.

1. Fill in the blanks with the terms provided.

mean	range	median	interquartile range(IQR)

The *mean* is the arithmetic average of a set of numbers. It is a measure of center, or central tendency.

The *range* is the difference between the highest data value and the lowest data value in a data set. It is a measure of variation.

A measure of variation in a set of numerical data, the *interquartile range (IQR)* is the distance between the first and third quartiles of the data set.

The *median* is the middle of an ordered list of the values. If the list has an odd number of values, it is the middle value of that list. If the list has an even number of values, it is the average of the two middle values. It is a measure of center, or central tendency.

Suppose an office tracks the number of phone calls received on business days, Monday through Friday, over two weeks. The data set is shown below.

183	165	58	152	179
152	153	135	124	112

2. Discuss with your partner what you notice about the data set.
Answers may vary. There are two 152's. Most numbers are in the hundreds. Fifty-eight looks out of place.

3. Match the measures of center and variation with their calculations for the data set above.

A. Mean	B. Median	C. Range	D. IQR

B	$\frac{152+152}{2} = 152$
D	$165 - 124 = 41$
A	$\frac{183+165+58+152+179+152+153+135+124+112}{10} = 141.3$
C	$183 - 58 = 125$

13.1.2 REFRESHER: MEASURES OF CENTER AND VARIATION

DIFFERENTIATE INSTRUCTION

Consider displaying a graph of data, such as a dot plot, if students are struggling with the idea of center and variation. Visualizing the data may help students understand what center and variation mean. "Variation" may be a difficult vocabulary word for some students. Using synonyms such as "spread" or "dispersion" may help.

TEACHER MOVES

This Refresher is vocabulary heavy to activate students' prior knowledge of these terms. Students learned these terms in Grade 6 and should be familiar with them. Determine how much of the Refresher is necessary for review based on student responses.

TEACHER MOVES

Notice and Wonder

To position students to begin to conceptually consider the concept of outliers, which will be reviewed at the end of this component, engage students in a Notice and Wonder routine. Ask questions like:

- What do you notice about the data set?
- What do the numbers have in common?
- What is different?
- What does this make you wonder?

13.1.2 REFRESHER: MEASURES OF CENTER AND VARIATION (CONTINUED)

ENRICHMENT

Students represented data using these graphs in 6th grade. Activate prior knowledge through questioning:

- What are the components of a line plot? Histogram? Box plot?
- What do you remember about each of these data displays?
- Which measures of center and variation can be determined from each display?

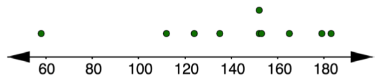
TEACHER MOVES

Students do not need to know how to determine whether a number is an outlier in Grade 7, so the values used as potential outliers have already been confirmed in these lessons. Consider explaining to students that there is a calculation to determine if a value in a data set is actually an outlier. They should not just assume any number larger or smaller than the others in a data set is automatically an outlier, though the examples they'll see in this unit are.

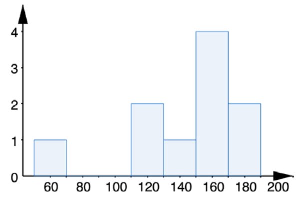
For the teacher's reference, a number is an outlier if it is 1.5×IQR above Q3 or below Q1 in a set of data.

4. A ***histogram***, ***line plot***, and ***box plot*** representing the data are shown.

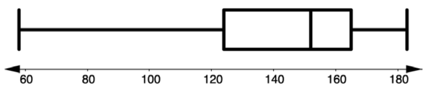
Line Plot



Histogram



Box Plot



a. Label the display types histogram, box plot, or line plot by filling in the box above each display.

b. Complete the table by identifying which measures of center and variation can be determined from each display.

	Mean	Median	Range	IQR
Histogram	☐	☐	☐	☐
Line Plot	■	■	■	■
Box Plot	☐	■	■	■

5. Are there any data values that are far away from the rest? 58

An ***outlier*** is a value that is much higher or much lower than the other values in a set of data.

13.1.3 Guided Instruction: Stem-and-Leaf Plots

Recall the data collected on the number of phone calls an office received on business days over two weeks.

183	165	58	152	179
152	153	135	124	112

In addition to displaying the data graphically with a histogram, line plot, or box plot, the data can also be displayed in a ***stem-and-leaf*** plot.

A ***stem-and-leaf plot*** is a table that organizes data by place value to compare data frequencies.

1. What do you notice about the data in the stem-and-leaf plot?
Answers will vary. There is a large gap in the data. There is a number that repeats, 152.

Stem	Leaf
5	8
6	
7	
8	
9	
10	
11	2
12	4
13	5
14	
15	2, 2, 3
16	5
17	9
18	3

Key: 5|8 = 58

2. Compare the stem-and-leaf plot with the line plot from the refresher.

What similarities do you notice between the displays?	What differences do you notice between the displays?
Answers will vary. Both displays show a gap in the data.	*Answers will vary. The stem-and-leaf plot shows exact values. The line plot's scale makes it hard to determine what the value of each data point is from the display.*

3. What value does the entry 13|5 represent? 135

4. Which measure(s) of center can be determined from the stem-and-leaf plot?
   - ○ mean
   - ○ median
   - ○ both the mean and median

5. Which measure(s) of variation can be determined from the stem-and-leaf plot?
   - ○ range
   - ○ interquartile range
   - ○ both the range and the interquartile range

13.1.3 GUIDED INSTRUCTION: STEM-AND-LEAF PLOTS

TEACHER MOVES

Notice and Wonder

Review the definition of stem-and-leaf plot with students, then use the plot in question 1 to draw students' attention to the features of stem-and-leaf plots. Engage students in a Notice and Wonder routine and circulate as they discuss in groups or pairs. Students may connect the stem-and-leaf plot to a vertical number line where each ten, in this case, is marked, but there are only values in the second column when there is an actual data value from the set that falls within that interval of 10. Accounting for each interval of 10 within the range allows the display to show the spread of the data.

DIFFERENTIATE INSTRUCTION

Stem-and-leaf plots are a useful tool. Students can visually see the large gap in the data, meaning outliers can be more easily identified. Consider creating an anchor chart with students that identifies the parts of stem-and-leaf plots for students to reference throughout this unit.

ENRICHMENT

Display both plots for students to see. Have students work with a partner to discuss the similarities and differences between the displays. As students compare the displays, circulate and ask questions like:

- What do you notice about the values on each display? What does this mean for determining the measures of center and variation?
- What do you notice about the scale of the line plot? Why is this important?

ENRICHMENT

What does it mean when there are no values in the leaf column for some of the stems?

13.1.4 COLLABORATION: DETERMINING, DESCRIBING, AND COMPARING NUMERICAL DATA REPRESENTED GRAPHICALLY

DIFFERENTIATE INSTRUCTION

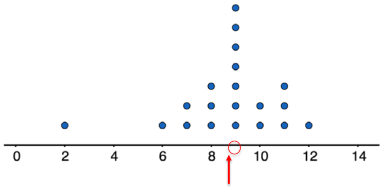
Auditory entry points are built into this component through partner discussion. Intentionally pair students of varying abilities together, particularly keeping in mind reading ability, as this component is reading-comprehension heavy.

13.1.4 Collaboration: Determining, Describing, and Comparing Numerical Data Represented Graphically

Work with your partner to complete the following.

1. Twenty students timed how long it took each of them to solve a puzzle. Below is a list of their solution times, in minutes, and a line plot displaying the data.

$2, 6, 7, 7, 8, 8, 8, 9, 9, 9, 9, 9, 9, 9, 10, 10, 11, 11, 11, 12$



Puzzle Solving Time (minutes)

a. Determine the mean of the data set. Then, draw an arrow (↑) below the number line that points to this value.
Mean = 8.7
b. Determine the median of the data set. Then, circle the value on the number line that represents the median.
Median = 9
c. Discuss with your partner what you notice about the mean and the median of the data. Summarize your discussion.
Answers will vary. The mean and the median are very close together. They only differ by 0.3.
d. What is the range of the data? 10
e. Find the interquartile range (IQR).

Q1	Q3	IQR
8	10	2

2. Imagine that the fastest time of 2 minutes was removed from the data set. Discuss with your partner what effect, if any, removing this value has on the values of center and variation (mean, median, range, IQR). Summarize your discussion.
Answers will vary. The mean increases to approximately 9.05, the range decreases to 6, but the other measures do not change.

3. Describe how the shape of the graph would change if 2 is removed.
Answers will vary. There would no longer be a tail on the left side of the line plot. The shape would be more symmetric.

4. Complete the table to compare the values of each measure before and after 2 is removed.

	Original Value	>, <, =	New Value After ▯ Is Removed
Mean	8.7	<	9.05
Median	9	=	9
Range	10	>	6
IQR	2	=	2

5. Which measure(s) of center will be affected if the outlier is removed?
   - ○ mean
   - ○ median
   - ○ both the mean and median

6. Which measure(s) of variation will be affected if the outlier is removed?
   - ○ range
   - ○ interquartile range
   - ○ both the range and the interquartile range

13.1.4 COLLABORATION: DETERMINING, DESCRIBING, AND COMPARING NUMERICAL DATA REPRESENTED GRAPHICALLY (CONTINUED)

ENRICHMENT

This component is intentionally scaffolded to build toward student understanding of how outliers impact the measures of center and variation. If you notice students struggling with questions 2 and 3, further scaffold through questioning:

- Do you think the mean will increase or decrease if the lowest value, 2, is removed? Why?
- What will happen to the median if the lowest value is removed? Will it change a lot? Why or why not? How can you use the line plot to help you?
- How will the range change if the lowest value is removed? Will it be larger or smaller? Why?

QUESTIONING

For students who may finish early, challenge them to predict what would happen to a data set if an outlier that was much larger than the rest of the data set was removed. How would this impact the mean and median? How would it impact the measures of variation? Why?

TEACHER MOVES

A misconception that may arise is students may think all the measures of center and variation will be affected if the outlier is removed. Encourage students to refer back to the visual representations of the data, like line plots, and have them cover up the outliers. Have them consider how the display changes without the outlier(s), specifically focusing on the median. Guide students to seeing how the median will generally remain in the same spot, meaning it will not change considerably.

13.1.5 YOUR TURN!

QUESTIONING

To challenge students and further build conceptual understanding, ask questions like:

- What are some reasons statisticians may remove an outlier?
- What are some possible reasons outliers could have occurred in this context?

13.1.5 Your Turn!

The following data set shows the number of tickets a movie theater sold in a day for each movie.

162	623	76	277	181	64	105
265	82	263	124	167	150	245

1. One of the values in the data set is an outlier. Explain which value you think is the outlier.
   Answers will vary. 623
   This value is much larger than the rest of the data set.

2. Complete the statement.

   Removing the outlier, 623, causes the mean to
   - ○ increase
   - ○ decrease
   - ○ remain the same

   and the range to
   - ○ increase
   - ○ decrease
   - ○ remain the same

   because it is much
   - ○ larger
   - ○ smaller

   than the other values in the data set.

13.1.6 Wrap-Up: Reflection

1. Rate yourself on today's learning target using the scale below.
   Answers will vary.

   I can determine the implications an outlier has in a given data set.

2. Write an explanation of why you rated yourself where you did.
   Answers will vary.

13.1.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.